



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
12-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

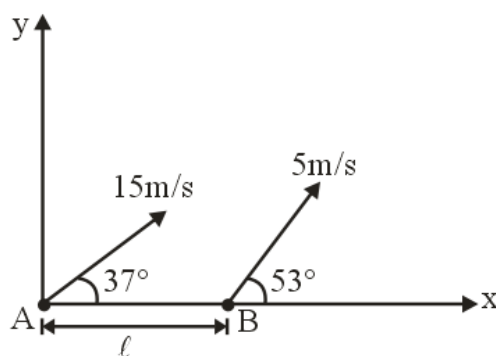
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Two spherical bodies A and B of same density and radius in the ratio 1 : 2 are projected simultaneously in xy-plane with velocities 15 m/s and 5 m/s as shown. Force on bodies is directly proportional to their radius and directed parallel to negative y-axis. Acceleration of A is given as $-4\hat{j} \text{ m/s}^2$. Find distance ℓ if they collide.



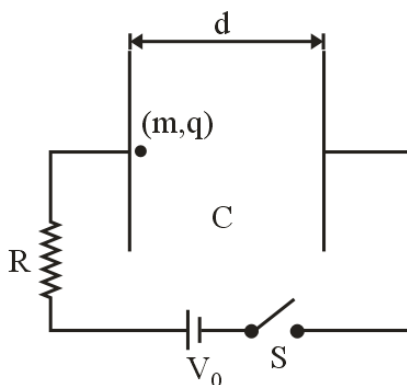
- (A) 10 m (B) 20 m (C) 30 m (D) 40 m
2. A small spherical smooth body of mass 1 kg moving with velocity $(2\hat{i} - 3\hat{j}) \text{ m/s}$ collides freely with another spherical smooth body of mass 2 kg at rest and rebounds with a velocity of $(\hat{i} + \hat{j}) \text{ m/s}$. Maximum kinetic energy loss in collision is :

- (A) $\frac{13}{3} \text{ J}$ (B) $\frac{89}{34} \text{ J}$ (C) $\frac{11}{3} \text{ J}$ (D) $\frac{196}{51} \text{ J}$

Space for Rough Work

3. In the given figure symbols have their usual meanings. Capacitor is uncharged initially. At time $t = 0$, switch is turned on and a particle of mass m and charge q is released from rest just near the left plate. Assuming gravity and induction effect is negligible. (i.e. force on charge particle due to induced magnetic field is negligible and electric field of capacitor is unaffected by the charge particle) For

$d > RC \sqrt{\frac{qV_0}{m} \left(1 - \frac{1}{e}\right)}$, velocity of particle at time $t = RC$ will be :



- (A) $\frac{qV_0RC}{md} \left(1 - \frac{1}{e}\right)$ (B) $\frac{qV_0RC}{md}$
 (C) $\frac{qV_0RC}{mde}$ (D) $\frac{qV_0RC}{md} \left(1 + \frac{1}{e}\right)$
4. In Rutherford's α -particle experiment distance of closest approach is ℓ when impact parameter is zero. If for α -particles of same kinetic energy impact parameter be $\sqrt{2}\ell$ then distance of closest approach will be :
- (A) ℓ (B) $\sqrt{2}\ell$ (C) 2ℓ (D) 4ℓ

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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5. Pitch of a screwgauge is 2 mm and linear scale division is 1 mm. There are 100 divisions on circular scale. When thickness of an object is measured, 4th division is visible and 5th division is not visible. 70th division of circular scale is coinciding the reference line. If screwgauge has no zero error then

(A) Least count is 0.01 mm

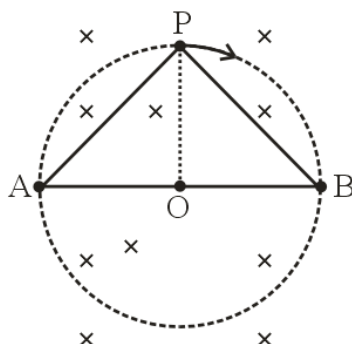
(B) Least count is 0.02 mm

(C) Thickness of object is 5.4 mm

(D) Thickness of object is 4.4 mm

Space for Rough Work

6. In the diagram. AB is a fixed conducting rod. PA and PB are conducting stretchable strings of negligible natural lengths. P moves on a semicircular path of radius R with constant speed V from A to B as shown. A uniform magnetic field B exists perpendicular to the plane of circle. If $\mathcal{E}$ be the induced e.m.f. in loop APBA then



- (A) $|\mathcal{E}|$ first increases then decreases
 (B) $|\mathcal{E}|$ first decreases then increases
 (C) $|\mathcal{E}| = \left| BRV \cos\left(\frac{Vt}{R}\right) \right|$
 (D) $|\mathcal{E}| = \left| BRV \sin\left(\frac{Vt}{R}\right) \right|$

Space for Rough Work

7. A uniformly charged disc with surface charge density of σ is rotating at a constant rate of ω . Given that radius of disc is R . Take all units to be SI. Choose the correct answer.
- (A) Ratio of magnetic induction at centre to magnetic induction on axis at distance R from centre is $\frac{\sqrt{2}}{3 - 2\sqrt{2}}$
- (B) Ratio of magnetic induction at centre to magnetic induction on axis at distance R from centre is $\left(\frac{3}{2\sqrt{2}} - 1\right)$
- (C) increase of 1% in speed of rotation of disc would cause about 2% increase in magnetic induction at a point on its axis
- (D) Very close to the center, magnetic induction $B \propto R$, where R is radius of disc.
8. A thin uniform rod of length L and specific gravity $\frac{2}{3}$ is hinged at one end to a point $\frac{L}{3}$ above surface of a liquid of specific gravity $\frac{3}{2}$. In case of equilibrium, choose correct option/options.
- (A) 74.5% portion of rod is outside of liquid.
- (B) $\left(\frac{3 - \sqrt{5}}{3}\right) L$ length of rod is immersed in liquid.
- (C) inclination of rod with surface of liquid is about $\sin^{-1}(0.45)$
- (D) Apparent weight of rod is 42.7% of its weight in air.

Space for Rough Work

9. In the red giant state of a star its surface temperature is 2000 K and in its white dwarf state its surface temperature, is about 20,000 K. Consider the size (volume) of white dwarf state to be 10^{15} times smaller than that of red giant. Take every state to be behaving as perfect black body and diameter of white dwarf to be 4000 Km. Choose correct option/options.
- (A) Total power radiated by star in its red giant state is about 1.14×10^{20} W
- (B) The average power radiated per unit area by white dwarf is about 10^4 times the average power radiated per unit area by red giant.
- (C) A medium size star converts into a white dwarf in its last days because expansion due to nuclear forces go weaker compared to gravitational contraction in the core of star.
- (D) Total power radiated by star in W.D. state is about $1.4\pi \times 10^{23}$ W
10. A ring of radius R and mass M is placed on a rectangular plank of same mass. Whole system is placed on a horizontal smooth floor. Plank is being pulled by external force F towards right. If ring does not slip over the surface of plank. Choose correct option/options.
- (A) linear acceleration of plank is $\frac{2F}{3M}$
- (B) linear acceleration of ring is $\frac{3F}{2M}$
- (C) Ratio of linear acceleration of ring to linear acceleration of plank is 0.5.
- (D) Friction working on ring is $\frac{2F}{3}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

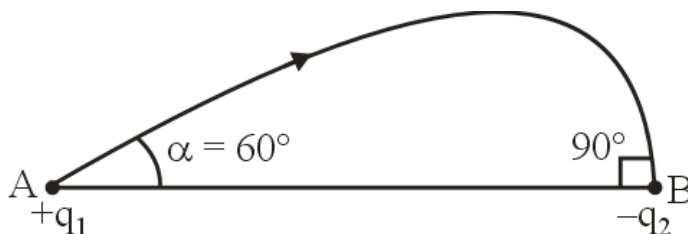
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Two point charges $+q_1$ and $-q_2$ are placed at A and B respectively. An electric line of force emerges from q_1 making an angle $\alpha = 60^\circ$ with line AB and terminates at $-q_2$ making an angle of 90° with the line AB.

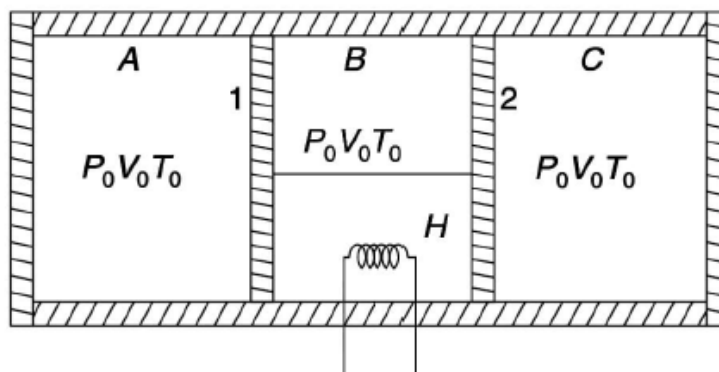


1. Find $\left| \frac{q_1}{q_2} \right|$
2. Find the maximum value of angle α at which a line emitted from q_1 terminates on charge q_2 . [In degrees]

Space for Rough Work

Paragraph for Questions 3 and 4

An insulated cylinder is divided into three parts A, B and C. Pistons 1 and 2 are connected by a rigid rod and can slide without friction inside the cylinder. Piston 1 is perfectly conducting while piston 2 is perfectly insulating. Equal quantity of an ideal gas is filled in three compartments and the state of gas in every part is same ($P_0 V_0 T_0$). Adiabatic exponent of the gas is $\gamma = 1.5$. The compartment B is slowly given heat through a heater H such that the final volume of gas in part C becomes $\frac{4V_0}{9}$

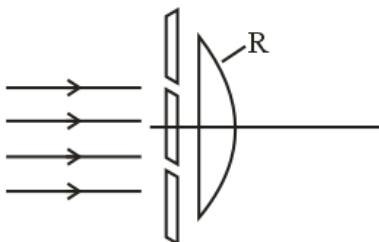


3. If the heat supplied by the heater is $XP_0 V_0$ then fill the value of X.
4. If the amount of heat flow through piston 1 is $YP_0 V_0$ then fill the value of Y.

Space for Rough Work

Paragraph for Questions 5 and 6

A screen with two slits at d distance apart are placed in path of parallel light beam. A plane convex lens is placed just after the slits. When light is observed in focal plane of lens interference fringes are observed.



5. If focal length of lens is 20 cm & $d = 1\text{mm}$, fringe width obtained with light of wavelength $\lambda = 5000 \text{ \AA}$; in mm is
6. If light was non-monochromatic having wavelength $4000\text{\AA}$ & $5000\text{\AA}$ what will distance (in mm) in focal plane of lens between point's of complete darkness. Assume no chromatic aberration in lens.

Space for Rough Work

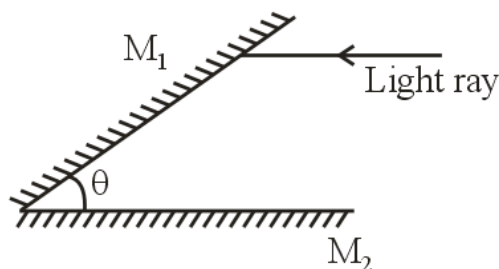
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A light ray is parallel to Mirror M_2 is incident on mirror M_1 . If ray refraces it's path after 10 reflection, then angle θ (in degree) between two mirrors is.



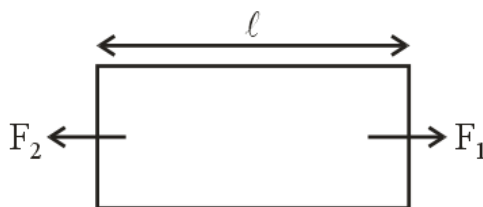
Space for Rough Work

8. A simple pendulum of length ℓ is oscillating with angular amplitude of 1° . If we do not consider approximation ($\sin \theta \simeq \theta$), then time period of oscillation is

- (1) Less than $2\pi\sqrt{\frac{\ell}{g}}$
 (2) Equal to $2\pi\sqrt{\frac{\ell}{g}}$
 (3) More than $2\pi\sqrt{\frac{\ell}{g}}$

Put correct order number of statement 1, 2, or 3 as answer.

9. Force of 100N and 80N are applied on the opposite faces of a an elastic rod of young's modulus of elasticity $Y = 1.5 \times 10^{11} \text{ N/m}^2$. Given that rod is placed on a smooth horizontal surface, cross-sectional area of rod is 1mm^2 . Change in length of rod is 0.12 cm, length of rod in metre is



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

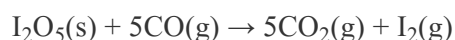
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1. The CO in a 20L sample of gas was converted to CO₂ by passing the gas over iodine pentoxide heated to 150°C



The iodine distilled at this temperature and was collected in an absorber containing 10 mL of 0.011 M Na₂S₂O₃. The excess hypo was back-titrated with 5 mL of 0.001 M I₂ solution. The milligrams of CO in 20 L of the original gas sample was therefore (I = 127)

- (A) 7.0 mg (B) 3.5 mg (C) 2.5 mg (D) 5 mg

2. Select the correct statement :

(A) A mixture of ideal gas is cooled upto liquid helium temperature (4.22 K) to form an ideal liquid solution

(B) In the following van der Waal's equation

$$\left[P + \frac{n^2 a}{V^2} \right] [V - nb] = nRT$$

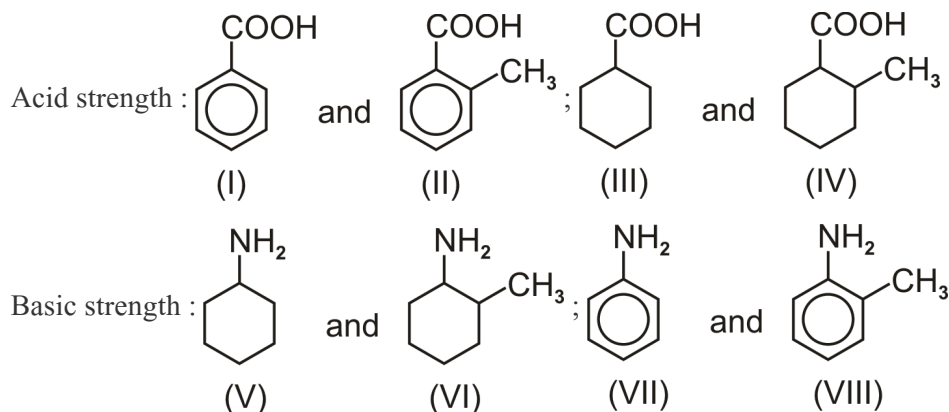
the constant 'a' reflects the strength of attractive forces

(C) A gas in a closed container will exert much higher pressure due to gravity at the bottom than at the top

(D) Kinetic energy of a molecule is zero at 0°C

Space for Rough Work

3. The correct order of acid and basic strength for the following pair of compounds should be ?



- (A) I > II ; III > IV ; V < VI ; VII < VIII (B) I < II ; III > IV ; V < VI ; VII > VIII
(C) I > II ; III > IV ; V > VI ; VII > VIII (D) I < II ; III > IV ; V < VI ; VII < VIII
4. Which of the following processes involves the concentration by leaching?

I - Bayer's Process

II- Separation of Cu^{+2} from its low grade ore

III - Cyanidation in Mac Arthur process

IV - Serpeck's process

- (A) I, II, IV (B) II, III, IV
(C) I, II, III (D) I, III, IV

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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5. Select the correct statement(s) according to Bohr's model

(A) Total energy = $\frac{-2\pi^2 K^2 Z^2 e^4 m}{n^2 h^2}$

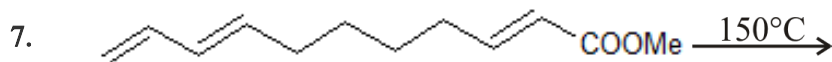
(B) Radius of orbit = $\frac{n^2 h^2}{4\pi^2 K Z e^2 m}$

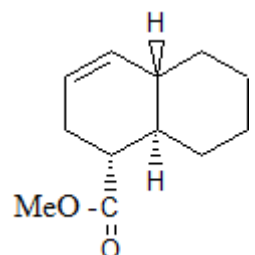
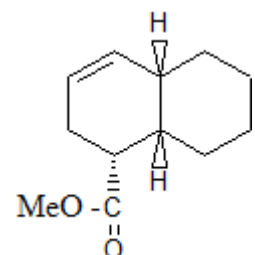
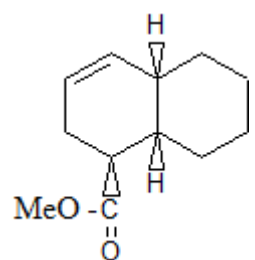
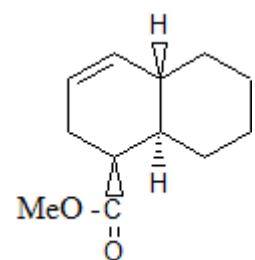
(C) Velocity of electron in an orbit = $\frac{2\pi K Z e^2}{nh}$

(D) Frequency of revolution of an electron in an orbit = $\frac{n^3 h^3}{4\pi K^2 Z^2 e^4 m}$

Space for Rough Work

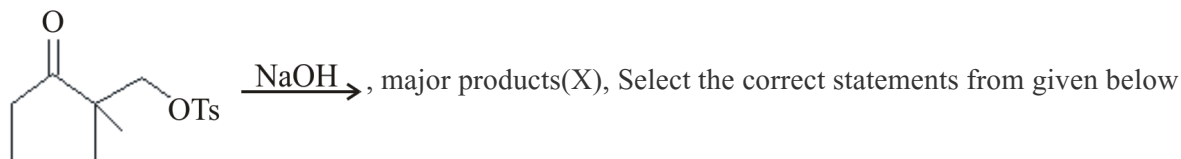
6. The dissociation of phosgene, which occurs according to the reaction $\text{COCl}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{Cl}_2(\text{g})$ is an endothermic process. Which of the following factors will increase the degree of dissociation of COCl_2 ?
- (A) Adding Cl_2 to the system
- (B) Adding helium to the system at constant pressure
- (C) Decreasing the temperature of the system
- (D) Reducing the total pressure



- (A) 
- (B) 
- (C) 
- (D) 

Space for Rough Work

8. Consider the reaction given below:



- (A) the major product gives effervescence with NaHCO₃ Soln.
 (B) (X) decolourises Br₂ water
 (C) formation of major product involves nucleophilic attack
 (D) The given reaction involves nucleophilic substitution
9. Identify the correct relationship of POCl₃ from the following
 (A) It is formed by the hydrolysis of PCl₅
 (B) It has equal bond length of all the bonds
 (C) the central atom is sp³ hybrid
 (D) It is the only product formed in the reaction of PCl₅ and P₄O₁₀
10. The thermal decomposition of compound gives CO₂ as the only gaseous product in
 (A) Washing Soda
 (B) Lime Stone
 (C) Quick Lime
 (D) Calcium Oxalate

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

An aqueous solution of a weak monoprotic acid is diluted from V_1 lit. to V_2 lit, Then change in pH is calculated as:



$$(I) \quad C_1(1-\alpha_1) \quad C_1\alpha_1 \quad C_1\alpha_1$$

$$(pH)_1 = -\log(C_1\alpha_1) \quad \text{..(i)}$$

$$(II) \quad C_2(1-\alpha_2) \quad C_2\alpha_2 \quad C_2\alpha_2$$

$$(pH)_2 = -\log(C_2\alpha_2) \quad \text{..(ii)}$$

$$\Delta pH = (pH)_2 - (pH)_1 = \log\left(\frac{C_1\alpha_1}{C_2\alpha_2}\right) = \log\left(\frac{\sqrt{K_a C_1}}{\sqrt{K_a C_2}}\right) = \frac{1}{2}\log\left(\frac{V_2}{V_1}\right)$$

Note: In both the cases degree of ionisation is very less than unity

1. 1 lit. of 0.1 M acetic acid is diluted in such a way that its pH increases by 0.5 unit then final volume (in lit.) will be
2. 1 lit. of 0.1 M aqueous solution of acetic acid is diluted in order to double the hydroxide ion conc. What is the final volume (in lit) of solution?

Space for Rough Work

Paragraph for Questions 3 and 4

An alkane(A), with the formula C_6H_{14} react with chlorine to yield three compound (with the formula $C_6H_{13}Cl$) B,C,D, of these only C and D can undergo dehydrohalogenation with sodium ethoxide in ethanol to produce an alkene. Moreover C and D yield the same alkene E (C_6H_{12}). Hydrogenation of E produces A. Treating E with HCl produces a compound (F) that is an isomer of B,C, and D. Treating F with Zn/CH_3COOH produces compound (G)

3. Number of B-hydrogen in 'E'
4. Number of primary hydrogen in 'G'

Space for Rough Work

Paragraph for Questions 5 and 6

Bent's Rule is related to the bond parameters in covalent molecules when lone pair and atoms of different electronegativity involve hybrid orbitals of different percentage of the contributing orbitals

5. How many of the following species involve the same percentage of s and p orbitals in each valence shell electron pair accommodation for the central atom?

CH_4 , NH_3 , NH_4^+ , PH_3 , PH_4^+ , POCl_3 , PCl_5 , SF_6 , ICl_3 , BF_3 , BF_4^- , XeF_2

6. How many  bond angles have value 90° in SF_6 ?

Space for Rough Work

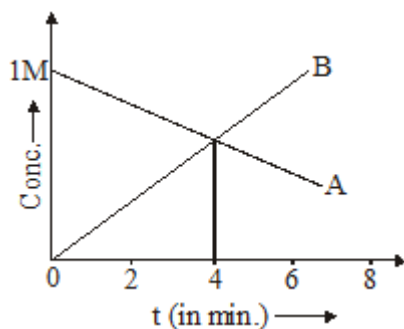
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

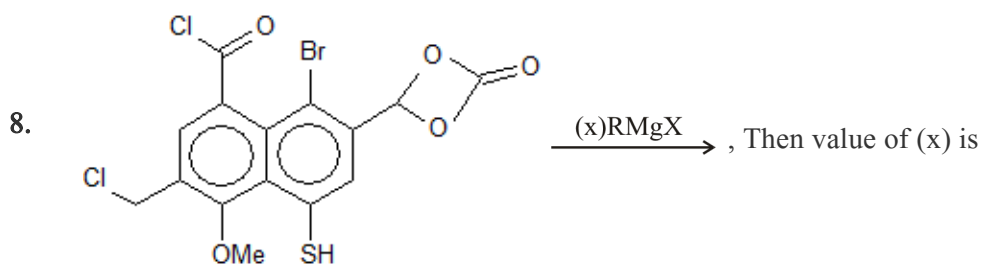
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. For zero order reaction $A \rightarrow 3B$ plot of concentration vs time is as follows:



Find half-life period in minutes if the initial conc. of A be 1M.



9. How many geometrical isomers are possible for the complex compound $[Co(en) Cl_2 Br_2]^-$?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If α and β are the roots of $x^2 + x + 1 = 0$ and $A_n = \alpha_n + \beta_n$, then AM of A_6 and A_7 is -
 (A) $-A_8$ (B) A_8 (C) $\frac{1}{2}A_8$ (D) $-\frac{1}{2}A_8$
2. If the sum of n terms of the series $\frac{5}{1.2.3^1} + \frac{7}{2.3.3^2} + \frac{9}{3.4.3^3} + \dots$ is S_n , then-
 (A) $S_3 = \frac{105}{108}$ (B) $S_n = 1 - \frac{1}{3^n \cdot n}$
 (C) $\lim_{n \rightarrow \infty} S_n = 1$ (D) $\lim_{n \rightarrow \infty} S_n = 3$

Space for Rough Work

3. Let a, b are natural numbers such that their LCM is 166320 and HCF is 792, then number of possible ordered pairs (a, b) is-
- (A) 8 (B) 16 (C) 32 (D) 64
4. Die A has 2 white and 4 red faces whereas die B has 4 white and 2 red faces. A coin is flipped once. If it shows a head, die A is thrown n times and if it shows a tail die B is thrown n times. If probability that die A is thrown when it is given that red turns up every time is $\frac{32}{33}$. Then the value of n is
- (A) 5 (B) 6 (C) 7 (D) 4
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Consider a variable line L which passes through the point of intersection P of the lines $3x + 4y = 12$ and $x + 2y - 5 = 0$ meeting the coordinate axes at points A and B. Then

(A) locus of middle point of segment AB has the equation $\frac{3}{y} + \frac{4}{x} = 4$.

(B) locus of the feet of the perpendicular from the origin on the variable line L has the equation $2(x^2 + y^2) = 4x + 3y$

(C) locus of centroid of the variable triangle OAB has the equation (where O is origin) $\frac{3}{y} + \frac{4}{x} = 6$.

(D) Locus of middle point of segment AB has equation $\frac{3}{y} - \frac{4}{x} = 4$

Space for Rough Work

6. Which of the following is(are) correct about parabola $(24x + 7y)^2 = 25(7x - 24y)$ -
- (A) Focus $\left(\frac{7}{100}, -\frac{6}{25}\right)$ (B) Directrix $7x - 24y + 25 = 0$
- (C) Length of latus rectum 1 (D) Vertex (0,0)
7. If $\sec^2\alpha + 6\cos^2\alpha = p$ & $\tan^2\alpha - 5\sin\alpha = p - 8$ where $\alpha \in \left(0, \frac{\pi}{2}\right)$, then which can be true-
- (A) $\operatorname{cosec}^2\alpha = 4$ (B) $\cot^2\alpha = 8$ (C) $p = \frac{35}{6}$ (D) $p = \frac{155}{24}$
8. If $(1+x)^{4n} + (1+x+x^2)^{2n} + (1+x+x^2+x^3+x^4)^n = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots + a_{4n}x^{4n}$, (where n is a natural number greater than or equal to 4), then -
- (A) $a_0 - a_1 + a_2 - a_3 + \dots + a_{4n} = 2$ (B) $a_0 - a_1 + a_2 - a_3 + \dots + a_{4n} = 2^n$
- (C) $\frac{a_r}{a_{4n-r}} = 1$ (D) $\frac{a_r}{a_{4n-r}} = \frac{a_{r-1}}{a_{4n-r+1}}$

Space for Rough Work

9. If from a point P representing the complex number z_1 on the curve $|z| = 2$, two tangents are drawn from P to the curve $|z| = 1$ meeting at points Q(z_2) and R(z_3); then
- (A) complex number $\frac{z_1 + z_2 + z_3}{3}$ will be on the curve $|z| = 1$
- (B) $\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right) \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) = 9$
- (C) $\left|\arg\left(\frac{z_2}{z_3}\right)\right| = \frac{\pi}{3}$
- (D) $|z_1 + z_2 - 2z_3| = 3$
10. Let P(r, s), ($r < 0$) is a variable point on ellipse $C_1 : \frac{x^2}{16} + \frac{y^2}{9} = 1$. PA and PB are two tangents of parabola $C_2 : y^2 - 4x = 0$, where A and B are point of contacts, then -
- (A) number of common tangents of curves C_1 and C_2 is 4.
- (B) minimum integral value of $r + s$ is -5
- (C) mid point of chord AB lie on curve $\frac{y^2}{9} + \frac{(y^2 - 2x)^2}{64} = 1$
- (D) mid point of chord AB lie on curve $\frac{x^2}{9} + \frac{(y^2 - 2x)^2}{64} = 1$

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider $f : [-1, 1] \rightarrow [3, 5]$ be a linear polynomial increasing function, then

1. $f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) =$
2. $f^{-1}\left(\frac{15}{4}\right) =$

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Paragraph for Questions 3 and 4

Consider a twice differentiable function $f(x)$ satisfying $f''(x) + f'(x) - 2f(x) = 0$ where $f(0) = 0$ and $f(1) = 3e$.

3. The number of zeroes of the equation $f(x) = 0$ is equal to
4. $\int_0^1 \left(-\frac{e^x x^3}{2} + f(x) \right) dx$ is equal to ke then k is (where 'e' is Napier's Constant)

Paragraph for Questions 5 and 6

Let $f(x) = x^3(x-1)(x-2)^2 \forall x \in \mathbb{R}$.

5. Least integral value of x at which $f(x)$ is increasing is -
6. Number of positive integral values of x at which $f(x)$ is decreasing is -
-

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SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If $a_n (n \geq 1)$ is a sequence of real number such that $a_1 = 12$ & $a_{n+1} = a_n \left(1 + \frac{3}{n+1}\right)$,
then if $L = \left(\lim_{r \rightarrow \infty} \sum_{k=1}^r \frac{1}{a_k}\right)$ find L^{-1}
8. If the area bounded by $y = x^2 + 6$ & $y = 4|x| + 2\cos(\pi x)$ is k , then the value of $\frac{3k}{20}$ is equal to
9. If $k = \int_{-2013}^{2013} \frac{dx}{1 + x^{2011} + \sqrt{1 + x^{4022}}}$ then the sum of digit of value k is

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work